

Force and Pressure

MCQ

1. Force is defined as a push or pull that can change the position, size, shape or speed of a body.

A) Energy
B) Force
C) Pressure
D) Power

Correct Answer: B

2. Force is a _____ quantity.

A) Scalar
B) Vector
C) Constant
D) Neutral

Correct Answer: B

3. The SI unit of force is

A) Joule
B) Pascal
C) Newton
D) Watt

Correct Answer: C

4. A car being pushed by a man is an example of

A) Magnetic force
B) Muscular force
C) Gravitational force
D) Electrostatic force

Correct Answer: B

5. Which of the following is NOT an effect of force?

A) Change in shape
B) Change in motion
C) Change in colour

D) Change in direction

Correct Answer: C

6. The force acting perpendicular to a surface is called

A) Pressure

B) Thrust

C) Weight

D) Friction

Correct Answer: B

7. Pressure is defined as

A) Force $\times$ Area

B) Area $\div$ Force

C) Force $\div$ Area

D) Mass $\div$ Volume

Correct Answer: C

8. The SI unit of pressure is

A) Newton

B) Joule

C) Pascal

D) Watt

Correct Answer: C

9. 1 Pascal is equal to

A) 1 N m²

B) 1 N/m²

C) 1 m/N²

D) 1 kg/m²

Correct Answer: B

10. When surface area decreases, pressure

A) Decreases

B) Remains same

C) Increases

D) Becomes zero

Correct Answer: C

11. Which state of matter has definite shape and volume?

- A) Solid
- B) Liquid
- C) Gas
- D) Plasma

Correct Answer: A

12. Liquids have

- A) Definite shape and volume
- B) No shape and no volume
- C) Definite volume but no shape
- D) No volume but shape

Correct Answer: C

13. Gases have

- A) Definite volume
- B) Definite shape
- C) No definite shape and volume
- D) Rigid structure

Correct Answer: C

14. The layer of air surrounding the Earth is called

- A) Hydrosphere
- B) Lithosphere
- C) Atmosphere
- D) Biosphere

Correct Answer: C

15. Atmospheric pressure is measured using

- A) Thermometer
- B) Barometer
- C) Hygrometer
- D) Ammeter

Correct Answer: B

16. The barometer was invented by

- A) Newton

- B) Pascal
- C) Torricelli
- D) Galileo

Correct Answer: C

17. Atmospheric pressure decreases with

- A) Temperature
- B) Depth
- C) Altitude
- D) Time

Correct Answer: C

18. 1 atm is equal to

- A) 760 cmHg
- B) 76 mmHg
- C) 760 mmHg
- D) 76 mHg

Correct Answer: C

19. 1 atm is approximately equal to

- A) 10^4 Pa
- B) 10^3 Pa
- C) 10^5 Pa
- D) 10^6 Pa

Correct Answer: C

20. The SI unit of atmospheric pressure is

- A) Newton
- B) Nm^{-2}
- C) Joule
- D) Watt

Correct Answer: B

21. The upward force exerted by a liquid on an object is called

- A) Thrust
- B) Friction
- C) Buoyant force

D) Pressure

Correct Answer: C

22. The phenomenon of floating or sinking is known as

A) Viscosity

B) Buoyancy

C) Density

D) Elasticity

Correct Answer: B

23. An object floats when

A) Weight > buoyant force

B) Weight = buoyant force

C) Weight < buoyant force

D) Weight = zero

Correct Answer: C

24. Liquids exert pressure

A) Only downward

B) Only upward

C) Only sideways

D) In all directions

Correct Answer: D

25. Pressure in liquids increases with

A) Area

B) Temperature

C) Depth

D) Density only

Correct Answer: C

26. The pressure exerted by a liquid at rest is called

A) Dynamic pressure

B) Atmospheric pressure

C) Static pressure

D) Vacuum pressure

Correct Answer: C

27. Pascal's law applies to

- A) Solids
- B) Gases only
- C) Confined liquids
- D) Open liquids

Correct Answer: C

28. According to Pascal's law, pressure is transmitted

- A) In one direction
- B) Unevenly
- C) Equally in all directions
- D) Only upward

Correct Answer: C

29. Hydraulic lift works on the principle of

- A) Buoyancy
- B) Pascal's law
- C) Archimedes' principle
- D) Friction

Correct Answer: B

30. Automobile brakes are based on

- A) Surface tension
- B) Pascal's law
- C) Viscosity
- D) Gravity

Correct Answer: B

31. Surface tension is due to

- A) Gravitational force
- B) Magnetic force
- C) Cohesive force
- D) Frictional force

Correct Answer: C

32. The SI unit of surface tension is

- A) Nm^{-1}

B) Nm^{-2}

C) N^2m

D) mN

Correct Answer: A

33. Insects walking on water is due to

A) Buoyancy

B) Pressure

C) Surface tension

D) Friction

Correct Answer: C

34. Floating of a needle on water is due to

A) Density

B) Pressure

C) Surface tension

D) Viscosity

Correct Answer: C

35. Viscosity refers to

A) Flow of solids

B) Resistance to flow of liquids

C) Expansion of gases

D) Compression of solids

Correct Answer: B

36. Viscous force opposes

A) Motion between layers

B) Gravity

C) Magnetism

D) Pressure

Correct Answer: A

37. The CGS unit of viscosity is

A) Pascal

B) Poise

C) Newton

D) Joule

Correct Answer: B

38. Friction is a force that

A) Increases motion

B) Stops gravity

C) Opposes motion

D) Changes mass

Correct Answer: C

39. Friction produces

A) Cold

B) Light

C) Heat

D) Sound

Correct Answer: C

40. Rubbing hands together is an example of

A) Static friction

B) Rolling friction

C) Sliding friction

D) Fluid friction

Correct Answer: C

41. Friction that acts on a body at rest is

A) Kinetic friction

B) Rolling friction

C) Static friction

D) Fluid friction

Correct Answer: C

42. A book kept on a table experiences

A) Kinetic friction

B) Rolling friction

C) Static friction

D) No friction

Correct Answer: C

43. Friction acting during motion is called

- A) Static friction
- B) Kinetic friction
- C) Limiting friction
- D) Zero friction

Correct Answer: B

44. Riding a bicycle is an example of

- A) Static friction
- B) Kinetic friction
- C) No friction
- D) Magnetic force

Correct Answer: B

45. Sliding friction occurs when

- A) Body rolls
- B) Body slides
- C) Body floats
- D) Body sinks

Correct Answer: B

46. Erasing with rubber involves

- A) Rolling friction
- B) Static friction
- C) Sliding friction
- D) Fluid friction

Correct Answer: C

47. Rolling friction occurs when

- A) Body slides
- B) Body rolls
- C) Body rests
- D) Body floats

Correct Answer: B

48. Rolling friction is _____ than sliding friction.

- A) Greater

B) Equal

C) Less

D) Zero

Correct Answer: C

49. Trolley tyres experience

A) Sliding friction

B) Rolling friction

C) Static friction

D) Fluid friction

Correct Answer: B

50. Friction depends on

A) Colour of surface

B) Shape of object

C) Nature of surface

D) Temperature only

Correct Answer: C

51. Friction increases when the surfaces are

A) Smooth

B) Polished

C) Rough

D) Lubricated

Correct Answer: C

52. Friction decreases when surfaces are

A) Rough

B) Pressed hard

C) Lubricated

D) Grooved

Correct Answer: C

53. Friction increases when the weight of the body

A) Decreases

B) Becomes zero

C) Increases

D) Remains same

Correct Answer: C

54. It is easier to ride a bicycle without load because

A) Air pressure is less

B) Friction is less

C) Speed is more

D) Gravity is less

Correct Answer: B

55. Friction depends on the

A) Colour of surface

B) Area of contact

C) Temperature only

D) Shape only

Correct Answer: B

56. Greater area of contact results in

A) Less friction

B) No friction

C) More friction

D) Zero force

Correct Answer: C

57. A road roller has a broad base to

A) Reduce pressure

B) Increase pressure

C) Increase friction

D) Reduce weight

Correct Answer: C

58. Friction between tyre and road helps in

A) Skidding

B) Slipping

C) Motion of vehicle

D) Stopping gravity

Correct Answer: C

59. Without friction, walking would be

A) Easier

- B) Faster
- C) Impossible
- D) Smooth

Correct Answer: C

60. Friction helps us to

- A) Fly
- B) Walk
- C) Float
- D) Sink

Correct Answer: B

61. Writing on paper is possible because of

- A) Gravity
- B) Pressure
- C) Friction
- D) Magnetism

Correct Answer: C

62. Friction between brakes and wheel helps to

- A) Increase speed
- B) Stop vehicle
- C) Reduce weight
- D) Increase fuel

Correct Answer: B

63. Friction is called a necessary evil because

- A) It is always harmful
- B) It is always useful
- C) It has both advantages and disadvantages
- D) It stops motion always

Correct Answer: C

64. Friction causes wear and tear of

- A) Air
- B) Water
- C) Surfaces in contact
- D) Light

Correct Answer: C

65. Friction reduces the _____ of machines.

- A) Weight
- B) Efficiency
- C) Speed
- D) Size

Correct Answer: B

66. Friction produces

- A) Cold
- B) Heat
- C) Electricity
- D) Magnetism

Correct Answer: B

67. Heat produced due to friction may

- A) Improve machines
- B) Damage machines
- C) Increase efficiency
- D) Reduce mass

Correct Answer: B

68. Soles of shoes are grooved to

- A) Reduce friction
- B) Increase friction
- C) Reduce pressure
- D) Increase speed

Correct Answer: B

69. Kabaddi players rub sand on hands to

- A) Reduce friction
- B) Increase friction
- C) Reduce weight
- D) Increase pressure

Correct Answer: B

70. Adhesive materials increase

- A) Pressure
- B) Speed
- C) Friction

D) Buoyancy

Correct Answer: C

71. Friction can be reduced by using

A) Sand

B) Rough surfaces

C) Lubricants

D) Adhesives

Correct Answer: C

72. Which of the following is a lubricant?

A) Sand

B) Water

C) Grease

D) Stone

Correct Answer: C

73. Lubricants make surfaces

A) Rough

B) Sticky

C) Smooth

D) Hard

Correct Answer: C

74. Ball bearings reduce friction by converting

A) Sliding into rolling

B) Rolling into sliding

C) Heat into energy

D) Motion into rest

Correct Answer: A

75. Rolling friction is reduced by

A) Lubricants

B) Ball bearings

C) Rough surfaces

D) Weight

Correct Answer: B

76. Roller skates use

A) Sliding friction

- B) Static friction
- C) Rolling friction
- D) Fluid friction

Correct Answer: C

77. Sprinkling powder on carrom board

- A) Increases friction
- B) Decreases friction
- C) Has no effect
- D) Increases weight

Correct Answer: B

78. Pressure is force acting

- A) Parallel to surface
- B) Tangential to surface
- C) Perpendicular to surface
- D) Along the surface

Correct Answer: C

79. Pressure depends on

- A) Force only
- B) Area only
- C) Force and area
- D) Mass only

Correct Answer: C

80. Sharp knives cut better because they exert

- A) Less pressure
- B) More pressure
- C) Equal pressure
- D) No pressure

Correct Answer: B

81. High-heeled shoes exert

- A) Less pressure
- B) More pressure
- C) Equal pressure
- D) No pressure

Correct Answer: B

82. Wide tyres are used in tractors to

- A) Increase pressure
- B) Reduce pressure
- C) Increase speed
- D) Reduce friction

Correct Answer: B

83. Atmospheric pressure is due to

- A) Weight of air
- B) Wind
- C) Temperature
- D) Clouds

Correct Answer: A

84. Atmospheric pressure is highest at

- A) Sea level
- B) Mountain top
- C) High altitude
- D) Space

Correct Answer: A

85. Buoyant force acts in

- A) Downward direction
- B) Sideways direction
- C) Upward direction
- D) Random direction

Correct Answer: C

86. Object sinks when

- A) Buoyant force $>$ weight
- B) Buoyant force $=$ weight
- C) Weight $>$ buoyant force
- D) Density is zero

Correct Answer: C

87. Pressure in liquids acts

- A) Only downward
- B) Only upward
- C) Only sideways

D) In all directions

Correct Answer: D

88. Liquid pressure depends on

A) Depth

B) Shape of container

C) Colour of liquid

D) Temperature only

Correct Answer: A

89. Water flows fastest from the hole

A) At the top

B) In the middle

C) At the bottom

D) All same

Correct Answer: C

90. Pascal's law is used in

A) Thermometer

B) Barometer

C) Hydraulic press

D) Speedometer

Correct Answer: C

91. Hydraulic lift is used to

A) Measure pressure

B) Lift heavy vehicles

C) Reduce friction

D) Increase speed

Correct Answer: B

92. Surface tension acts on the

A) Bottom of liquid

B) Entire liquid

C) Surface of liquid

D) Container

Correct Answer: C

93. Surface tension depends on

A) Nature of liquid

- B) Colour of liquid
- C) Volume of liquid
- D) Shape of container

Correct Answer: A

94. Liquid droplets are spherical due to

- A) Gravity
- B) Pressure
- C) Surface tension
- D) Viscosity

Correct Answer: C

95. Viscosity is resistance to

- A) Motion of solids
- B) Flow of liquids
- C) Flow of gases only
- D) Motion of air

Correct Answer: B

96. Highly viscous liquid flows

- A) Very fast
- B) Slowly
- C) Instantly
- D) Randomly

Correct Answer: B

97. Honey has _____ viscosity than water.

- A) Less
- B) Equal
- C) More
- D) Zero

Correct Answer: C

98. SI unit of viscosity is

- A) Poise
- B) Pascal
- C) N s m^{-2}
- D) Joule

Correct Answer: C

99. Friction between nail and wall helps nail to

- A) Slip out
- B) Fall down
- C) Stay fixed
- D) Melt

Correct Answer: C

100. Cutting wood is possible due to

- A) Pressure only
- B) Heat only
- C) Friction
- D) Gravity

Correct Answer: C